

Advanced (Habanero)

Q1. Factor out the greatest common factor of $2x + 6$

- (A) $2(x + 3)$
- (B) $6(x + 2)$
- (C) $2x + 6$
- (D) $x(2 + 6)$
- (E) $3(2x + 2)$

Q2. Simplify the expression $9x^2 + 12x$ by factoring

- (A) $3x(3x + 4)$
- (B) $3x(3x - 4)$
- (C) $12x(x + 1)$
- (D) $3x(3x + 2)$
- (E) $9x(x + 2)$

Q3. What is the greatest common factor of $12x$ and $15y$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q4. Factor out the greatest common factor of $10x^2y + 15xy^2$

- (A) $5xy(x + 3y)$
- (B) $5xy(2x + 3y)$
- (C) $10x(y + 2y)$
- (D) $5x(2xy + 3y)$
- (E) $5(2x^2y + 3xy^2)$

Q5. Simplify the expression $18x^3 + 12x^2$ by factoring

- (A) $6x^2(3x + 2)$
- (B) $6x^2(3x - 2)$
- (C) $6x(3x^2 + 2x)$
- (D) $6x(3x^2 - 2x)$
- (E) $9x(2x^2 + 2)$

Q6. What is the greatest common factor of $20x$ and $25y$?

- (A) 1
- (B) 2
- (C) 5
- (D) 10
- (E) 20

Q7. Factor out the greatest common factor of $8x^3 + 12x^2 + 16x$

- (A) $4x(2x^2 + 3x + 4)$
- (B) $4x(2x^2 - 3x - 4)$
- (C) $8x(x^2 + 2x + 3)$
- (D) $4x(2x^2 + 3x)$
- (E) $8x(x^2 - 2x - 3)$

Q8. Simplify the expression $15x^2y + 20xy^2 + 25x$ by factoring

- (A) $5xy(3x + 4y + 5)$
- (B) $5(3x^2y + 4xy^2 + 5x)$
- (C) $5xy(3x - 4y - 5)$
- (D) $5(3x^2y - 4xy^2 - 5x)$
- (E) $5(3x^2 + 4xy + 5)$

Open Ended Questions (FRQ)

Q9. Factor out the greatest common factor of $24x^3y + 18x^2y^2$ and simplify

Q10. Simplify the expression $12x^4 + 15x^3y - 9x^2y^2$ by factoring

Chef's Tips

- Provide the necessary tools for students to complete the test
- Allow students to use a calculator for complex calculations
- Ensure students understand that the greatest common factor is the largest factor that divides all terms of an expression